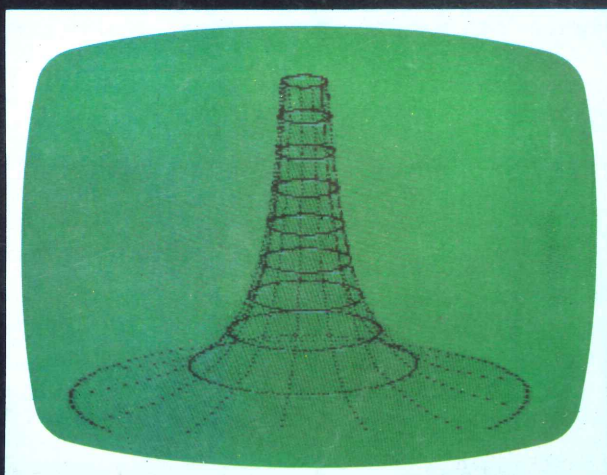
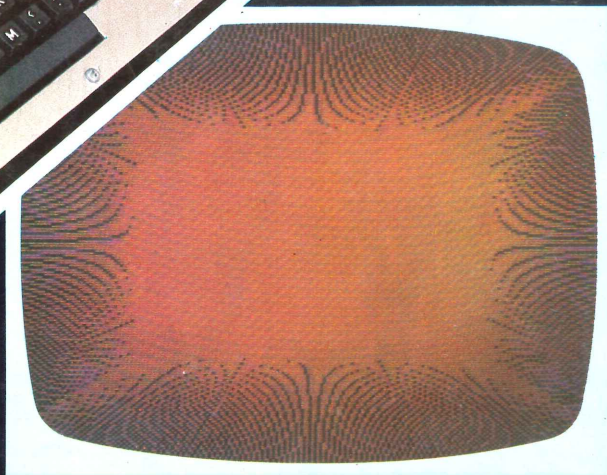
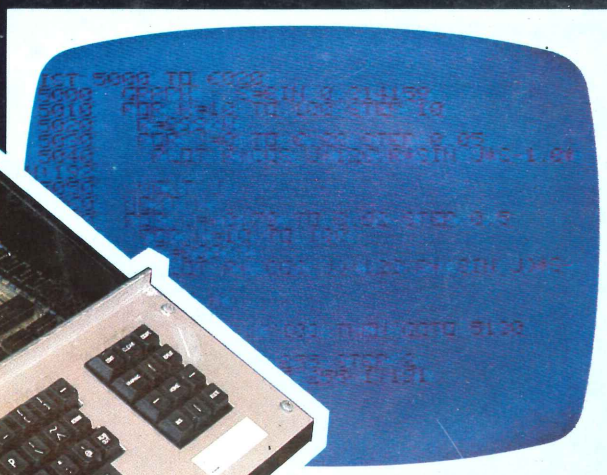


THE BIG ONE!

16-BIT COLOUR COMPUTER TO BUILD!



- Up to 16 serial I/O ports
- UCSD PASCAL on disc
- High resolution graphics
- Powerful resident BASIC
- Separate 16K video RAM
- Addressing up to 1M
- Up to 60K of user memory
- Floppy controller option
- Programmable baud rate



16 BIT COMPUTER

Announcing the ETI Cortex, in all its 16-bit splendour. This advanced design uses up-to-the-minute technology and forms the basis of a powerful home or business system.

Setting records is getting to be a habit at ETI. We were the first magazine to publish a DIY computer project (the System 68) and we were the first magazine to publish a one-board home computer (the Triton). Now we're the first magazine to publish a full-feature 16-bit computer, and at a price that makes a lot of commercial machines look a bit sick. The Cortex forms the basis of a versatile computer system that expands with your imagination and is based on state-of-the-art VLSI technology; the 'why use five chips if one will do?' philosophy.

Processing Power

The processor is a high-speed 16-bit device which possesses a unique system of RAM-based registers. The Cortex kit is supplied with a full 64K bytes of dynamic RAM (ie 32K words of 16 bits), and 24K of BASIC and assembler in an overlaid memory organisation (we'll explain what that means later). The high definition colour VDU has a separate 16K of RAM outside the CPU's 64K memory map and some extraordinary features that result in superb graphics capabilities. Disc drives may be interfaced easily as the controller chips fit on the PCB, and the resident BASIC can be overwritten by disc-based languages; the first that will be available is UCSD Pascal. The latter features will make the system particularly attractive to business users, and Cortexes (Corti?) will be available ready-built as well as in kit form.

Incidentally, we made a slip of the typewriter last month. What we meant to say was "Chips designed by Texas Instruments . . ." Sorry for any confusion caused.

The heart of the Cortex is the TMS9995 CPU. As with all the components in this project it was selected from the wide range of currently available CPUs on a price/performance basis. The 9995 is based on the unique memory-to-memory architecture of the TMS9900. Thus it has the same

architectural features of this powerful 16-bit processor and an enhancement of its rich, mini-computer style, instruction set. It is fabricated in state-of-the-art N-channel silicon gate MOS technology, enabling single 5V operation and high speed (12MHz) to be achieved in a compact silicon area.

As well as the 16-bit CPU, fabricated onto the same chip are a number of extra features that make this device the obvious choice for a large number of general purpose applications. 256 bytes (128 words) of on-chip RAM enables four complete, fast access register files (workspaces) to be implemented in full speed memory. A clock generator is also included to minimise the number of external components. Also available are a timer/event counter, a prioritised Interrupt Interface and a 16-bit flag register which can be output on the

I/O control bus. The I/O bus is completely separate from the main memory map, and enables 32K of individual I/O bits to be manipulated individually or simultaneously in groups.

Artful Architecture

The term 'memory-to-memory' implies the fundamental difference between traditional eight-bit CPU architectures and that of the 9995. That is, all transfers in the machine are from one main memory location to another. Only three 16-bit registers exist on the CPU itself; the Program Counter, the Status Register and the Workspace Pointer.

The Workspace Pointer contains the address of the start of a block of RAM anywhere in the main memory map. This address is designated register zero; the next 15 contiguous memory locations are designated Registers 1-15. These registers may then be used by the



A complete Cortex system.

programmer as scratch registers. A large number of instructions exist that make maximum use of the reduced addressing needed to access these. For example MPY R4, R5, performs a 16-bit multiply of the contents of Register 4 with Register 5 and stores the 32-bit result in Register 5 (most significant word) and Register 6.

What advantage does this give a programmer over other architectures? Well, in addition to providing no restriction on the choice of addressing modes (register 0 can still be accessed as memory location 7XXXX) the power comes when an interrupt or other subroutine call occurs. It is obviously desirable, especially on receipt of an interrupt, to be able to react as rapidly as possible. On a register-oriented machine it is necessary to save the contents of the working registers in main memory, a slow operation requiring a large number of reads and writes to push the registers onto the stack. On the 9995 the context switch occurs by changing the value in the Workspace Pointer. This points to a new area of fresh registers ready to process the interrupt. The full context of the previous operation is retained in the previous workspace registers, which, being in main memory, are still preserved intact. A return is similarly implemented

easily and rapidly by restoring the old Workspace Pointer value. In many real time control situations with a large number of external events occurring this architecture is the only one that allows for efficient processing.

External Affairs

The system clock is externally generated and fed to the CPU via the XTAL2 input. A clockout signal is also provided that is one-fourth of the crystal frequency (3MHz). The 64K bytes of system memory are directly addressed by A0-A15. Here the convention is that A0 is the most significant bit. A0-A14 are also used to directly address the separate I/O structure of the Communication Register Unit (CRU). The bit to be accessed is held on A0-A14 and the data is present on A15. The data is then clocked out by CRUCLK. Reading a CRU bit into the CPU is achieved in the same manner but is read into the CRUIN line.

The data bus is multiplexed from the internal 16-bit architecture to eight bits externally, D0-D7. A memory access is signalled by MEMEN and data direction is controlled by DBIN. WE indicates a memory write. For the control of slow memories a READY line signals to the CPU if the memory is ready to complete the current access. If not, the CPU waits for another

CLKOUT cycle before continuing.

In the Cortex all these features are used to provide the fastest BASIC available to a home constructor. Even then the speed is still limited by I/O transfers. This can be seen as evidenced by the amount of time the 'idle' LED is on. This LED is directly driven by a status decoder that indicates when the CPU is no longer executing any instructions but is waiting for an external interrupt.

Dynamic Designing

We don't doubt that all you digital dabblers have watched the plummeting price of memory in our advertisers' pages, and the most dramatic trend has been in dynamic memory (DRAM). This is partly due to the simplicity of the basic cell design in a DRAM; and this simplicity means in turn that the most dramatic cell density increases have been and will be made in DRAMs. With 16K DRAMs prices have pretty much levelled out at the bottom of the curve, but the 64K DRAMs are in hot pursuit.

More importantly from the technical point of view, certain 64K DRAMs offer upward compatibility to the next generation of 256K DRAMs. (Why are the steps quadrupling: 16K to 64K to 256K? Because DRAMs have a two-part

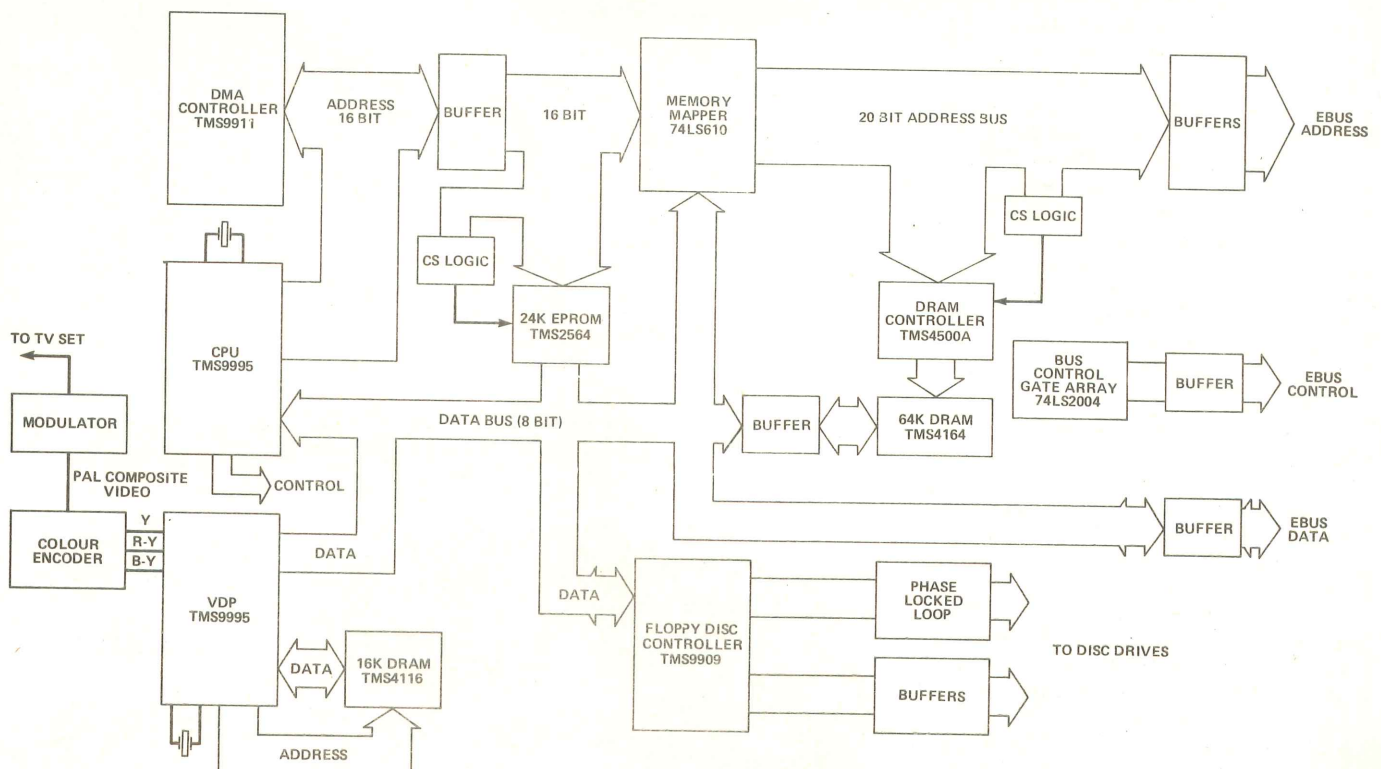


Fig. 1 Block diagram for the complete Cortex.

PROJECT: Cortex Computer

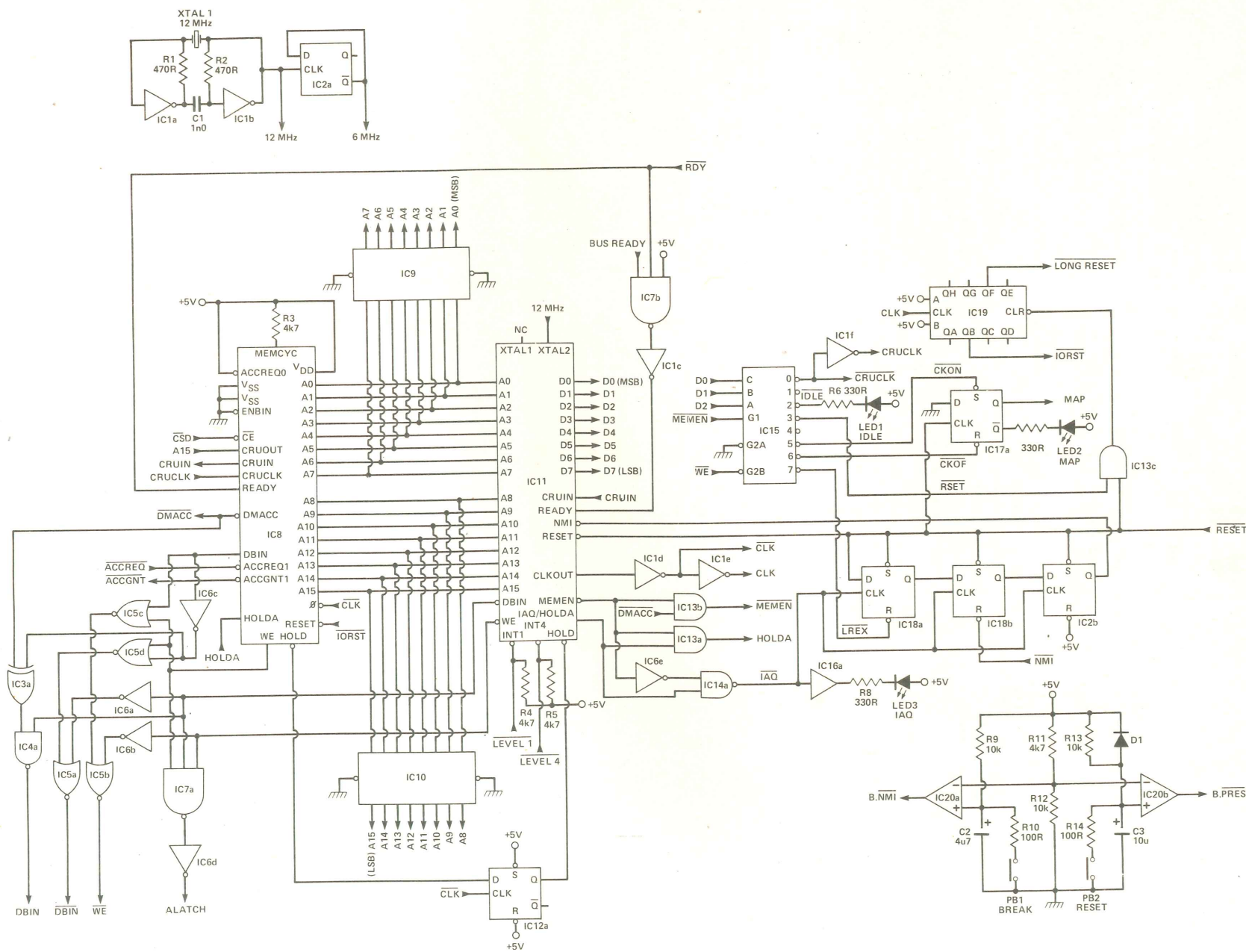


Fig. 2 Circuit diagram of the CPU and DMAC circuitry.

HOW IT WORKS — CPU AND DMAC

The heart of the system is the CPU, IC11 (a TMS9995). It has a 16-bit internal architecture and an eight-bit wide external data path. The master clock for the system is formed by IC1a,b and associated components; the 12Mhz clock rate of the CPU enables it to complete a memory read or write in only 166 nS! This is too fast for present DRAM technology so the automatic wait state feature of the CPU is used. This automatically assumes that memory is not ready and extends the memory access to 500 nS. The cycle can be further extended by a low level on the READY input to the CPU; this occurs, for example, when the DRAM is not ready because a refresh cycle is taking place.

The CPU signals the type of memory cycle by driving either DBIN or WE (write) low after driving MEMEN low. If the memory cycle is an instruction fetch then the IAQ/HOLDA signal goes high until both bytes have been fetched. This condition is decoded by IC6e, IC14a and buffered by IC16a to light LED3 to provide a front panel indication.

The CPU has a bit-mapped I/O interface which is separate from the memory data bus; the process is carried out by a section of the CPU called the Communications Register Unit (CRU). The data transfers are serial, bit by bit, each bit having a unique address. This allows 32K bits to be accessed (not 64K since address line A15 carries the data). The value of the data bit is on CRUOUT (A15) for output cycles and a WE/CRUCLK pulse is generated to strobe the data into the I/O devices. On input cycles the data is sampled from the CRUIN line and a pulse is generated on the DBIN line to enable the bus buffers and so on. During all serial I/O operations the MEMEN signal stays high. Any number of bits from one to 16 can be transferred, each bit taking 500 nS to transfer if the READY input is high.

There are some special I/O signals for control use called IDLE, LREX, CKON, CKOF and RSET. IC15 decodes these separately from the normal I/O operations by using the three-bit code output on D0-D2 of the data bus. IDLE pulses continuously whenever the IDLE instruction is executed; it indicates that the CPU is in an internal loop waiting for interrupts (ie doing nothing). LED1 on the front panel lights to indicate this state. LREX is used for the single instruc-

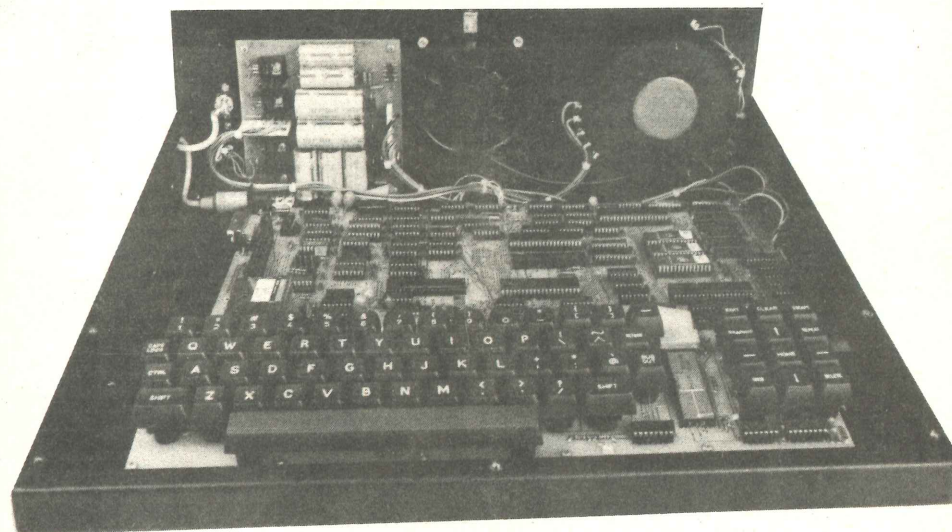
tion execution logic which causes an NMI interrupt to occur two instructions after executing the LREX instruction. The two-instruction delay is generated by the series flip-flops IC18a, IC18b and IC2b.

CKON and CKOF are used to switch the memory mapper device from passive to active and vice versa: the signals set or reset the Q output of IC17a to enable or disable the memory mapper (IC26) via IC24a, 25a. When Q is high, Q is low, and LED2 lights to signal that external memory is being accessed.

The RSET signal causes all I/O devices to be reset and sets the CPU interrupt mask to disable all interrupts. Normally both RESET and RSET are high, so the output of IC13c is high and IC19, an eight-bit parallel-out serial shift register, clocks out a continuous series of 1s. A low on RSET or RESET sets all the outputs of IC19 low (specifically IORST and LONG RESET); when the CLR input returns high, 1s are clocked through the shift register first taking IORST then LONG RESET high.

The Direct Memory Access controller (IC8, a TMS9911) is used to provide transparent high speed data transfer to and from the floppy disc controller (FDC) into memory. The address bus of the CPU is tri-state, as are the address outputs of the DMAC. Only one device is in control of the address bus at any one time. When the FDC requires the memory it signals on ACCREQ (access request); the DMAC then signals to the CPU using the HOLD signal that it requires the bus. When the CPU reaches the end of the current memory cycle it tri-states all its outputs (except MEMEN) and signals HOLDA ('acknowledged'). The DMAC now takes over the bus, signals ACCGNT to the FDC, and transfers the data byte between the memory and the FDC.

After completing the memory cycle(s) the DMAC then relinquishes control back to the CPU by releasing HOLD. The CPU then continues as if nothing had happened. The TMS9911 was designed for use with the TMS9900 CPU; when it is used with the TMS9995, extra gating is required to make the signals compatible. Parts of ICs 3, 4, 5, 6 and 7 take care of this. In this application only one of two channels in the DMAC is used; the other is free for the user to experiment with.



multiplexed address bus, so adding one extra address pin is the equivalent of two extra address lines and thus four times the addressing range.) Since all 16K DRAMs operated on a 128-cycle refresh, some 64Ks followed suit. However, that extra address pin called for twice the number of sense amplifiers on the chip, which occupied valuable silicon area (see this month's article on Designing Micro Systems for more about DRAM structure). For 256K DRAMs the waste of chip area is intolerable and to get a product that is capable of manufacture, 256-cycle refresh is essential. The TMS4164 64K DRAM, the first commercially available production device, adopted 256-cycle refresh from the start, as well as following the JEDEC-approved pin-out. This means that the devices are not only upward pin-compatible from the TMS4116, but will also be upward compatible in the future to 256K devices. Simply plug new chips in the old sockets and you've got four times the memory!

For these reasons we chose the TMS4164 to provide a full 64K

memory map using only eight chips. Not only is it compact, fast and very reliable, but it is the first 64K device to be successfully encapsulated in a low-cost plastic package.

In our application, refresh is achieved in a manner typical of the Cortex philosophy; a single-chip DRAM refresh controller is used, the TMS4500A. This device provides all the necessary control and arbitration functions for handling 64K DRAMs in a microprocessor system. It accepts the 16 address lines, A0-15, and multiplexes them to row address and column address (RAS and CAS) at the appropriate times, generates refresh signals for 128 or 256-cycle memories and arbitrates synchronously between access requests and refresh cycles. Synchronisation is important for achieving reliable arbitration, and the CPU clock is utilised as the main timing reference.

ROM To Manoeuvre

Similar design criteria were applied to the choice of EPROMs to store the system firmware. In order to economically store the large number of bytes required to provide

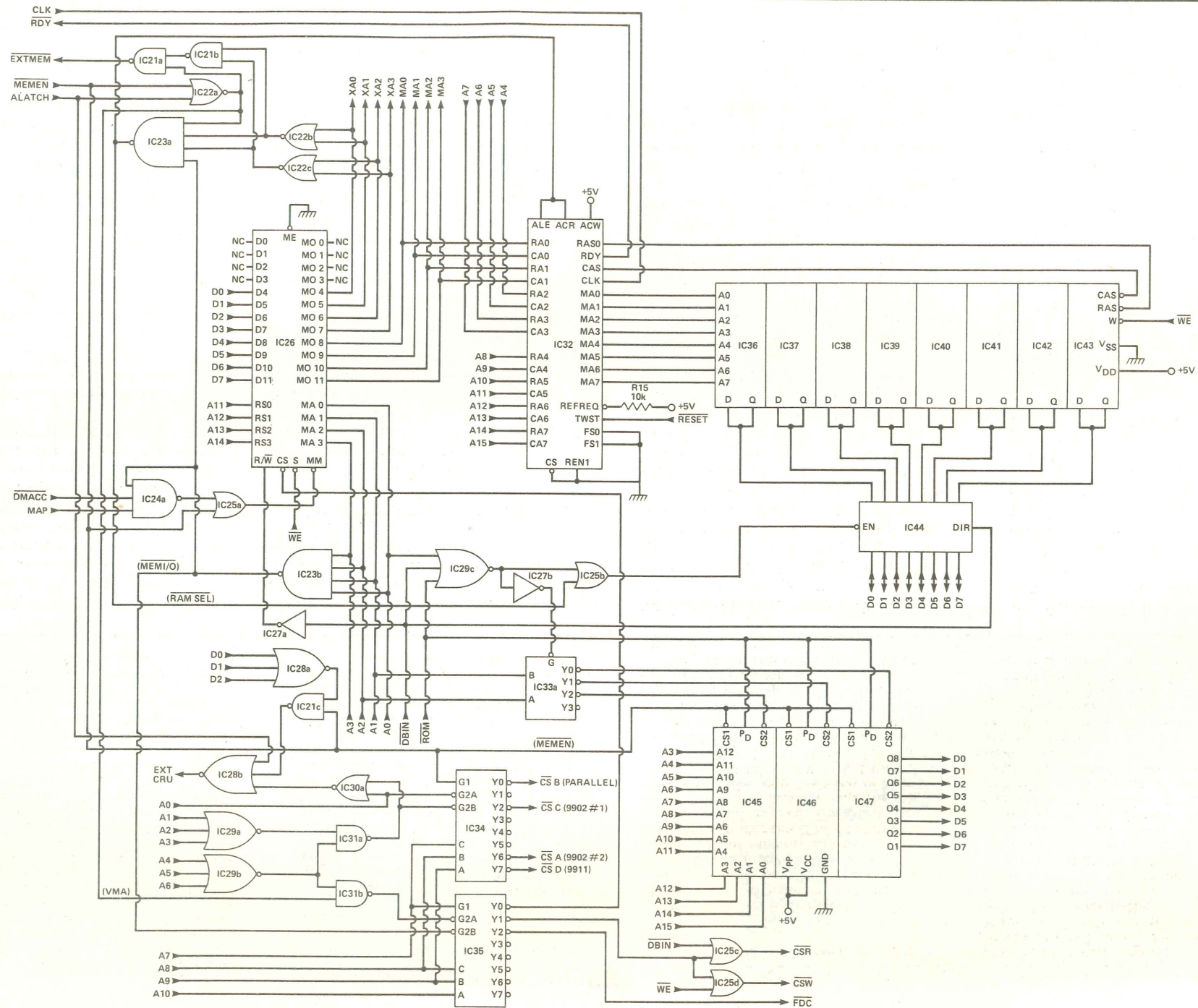


Fig. 3 Circuit diagram of the memory section.

The 24K of EPROM (IC45, 46, 47) contains the assembly language support and the BASIC interpreter. The EPROMs are switched in and out of the memory map by the I/O bit 'ROM' (see I/O section). This signal powers up in the active (low) state, with the EPROMs on. The DRAM (ICs 36-43) is also accessed during a read of the EPROMs but the data buffer IC44 is not enabled; this means that any write while the EPROMs are on is put into the DRAM, so that it behaves as a 'phantom' or ghost. The BASIC interpreter copies itself into the DRAM and then switches the EPROMs off. This has two advantages; first, during execution the interpreter overlays sections of code and then re-copies the relevant section of EPROM back to conserve memory. Second, to enable disc-based operating

systems (eg Pascal) to be used, the system needs to be able to operate RAM-resident.

The addresses for DRAM accesses are passed through the memory mapper device, IC26. This segments the CPU's 64K address map into 16 pages of 4K; each page has a 12 bit register (MO0 to MO11) of which only eight bits are used (MO4 to MO11). The outputs replace the top four bits from the CPU and add four more. Thus each 4K block can be anywhere in the 1M total address reach (20 bits). The CPU at any one time still only has a 64K address map but by dynamically loading the mapper during program execution the full 1M can be used. The mapper registers are loaded or read as 16 memory locations in the

memory-mapped I/O area in high memory.

The chip select logic defines the first 64K as on-board memory and the remaining memory map as off-board, using the E-BUS interface (see later). The bottom 32K of memory has the 'phantom' DRAM under the EPROM although only 24K of this is used. The DRAM occupies 60K of the memory map; the top 4K is sub-divided into 256 bytes of high-speed internal to the CPU, and then a memory-mapped I/O region for the Video Display Processor, the memory mapper and the floppy disc controller. Eight memory-mapped slots are decoded, leaving some spare chip selects for user experimentation and expansion.

The DRAM controller (IC32) takes

the 16-bit least significant address signals and multiplexes them on outputs MA0-7 to supply the memory devices with the correctly-timed waveforms. Once an access request is made, the addresses are first latched and then the controller arbitrates between a refresh cycle and an access cycle. If the controller is busy refreshing the memory then it signals a 'not ready' state to the CPU on the READY line, which suspends operation until the signal returns high again. The controller has to use 'cycle-steal' refresh, as the CPU is nearly always accessing memory and not enough free time can be guaranteed. The refresh cycle obviously slows down the CPU, but by less than 10%, which is a small penalty to pay for the large amount of memory at a low cost.

a comprehensive and versatile BASIC language, high capacity devices were required. The cost and capacity trends in EPROM technology have followed a similar path to DRAMs; thus the TMS2564 was chosen to complement the TMS4164 in the system. These devices, like all the major devices in the project, operate off a single 5V rail.

Each 2564 can store 8K of program organised in the now industry-standard 8K×8 format. Three chips are used to store the firmware: 24K in all. However, an important design feature of the Cortex should be mentioned here; the EPROMs phantom the DRAMs in the memory map. At power-up the EPROMs are enabled, and after checking the DRAMs the program then copies the full operating system from EPROM into RAM. Once this has been completed the EPROMs are disabled. Thus the operating system is running in RAM, allowing changes to be made or sections deleted to create extra space. This is most noticeable when you're only operating the assembler section, since the whole of the BASIC interpreter may then be eliminated from memory, freeing an

extra 20K for the user! Thus the Cortex is the only computer offering a full 64K of RAM to the user as a standard feature!

Graphic Descriptions

No home computer is considered worthy of the name these days unless it features impressive graphics capabilities, and the Cortex certainly won't disappoint you in this respect (just look at our cover and the photos in this article for some examples of the display power). What's more, all this is controlled with just one IC, the TMS9929! This chip is a memory-mapped video controller, which produces all the relevant video signals internally; it's a very versatile device capable of producing extremely complex displays, including 3D simulations. In order to achieve this its mode of operation is very different to the majority of display devices currently in use.

For simple text displays the video output of a computer may be produced by using a small random access memory to address a larger read-only memory that stores the character shape. Teletext displays fall into this category. The next step

up from this is to have a display where each displayed dot is a bit (or bit pattern) stored in RAM. The first method means that shapes can be positioned very rapidly but the repertoire of shapes is limited to those in the graphics/text ROM. The second scheme is slower but allows more complex shapes to be created and lines to be plotted. However, problems occur when trying to overlay shapes as everything must be done in software — making things even slower.

The 9929 uses variants on both these schemes to produce extremely complex shapes with the minimum of software overhead. Tables are designated in an area of RAM to define pattern shapes, characters or graphics. A separate table area is designated to define attributes to the shape, such as colour and size. Finally an area of RAM akin to the Teletext RAM contains pointers for each screen location that point to the desired shape and its associated attributes. The advantage of this system is that large, dramatic changes can be made to the display very rapidly by making simple changes in the pattern look-up table. Further, all displays of one particular shape can be modified

simultaneously by updating the pattern generator table.

A distinction is made in hardware between graphics and text modes, though text is available in graphics mode. They may be interchanged using the BASIC commands TEXT and GRAPH. In GRAPH mode the resolution is 256×192 (48K pixels), and the colour of each pixel can be set to any of the 16 available display colours, the only limitation being that only two colours per eight pixels may be used horizontally. To put it another way, the background may be any one of the 16 colours, while the colours of the pixels (foreground) can be set to one of 16 colours in blocks of 32×192. Control is exercised from BASIC using the COLOUR and PLOT commands.

In TEXT mode the screen format is 24 rows of 40 characters per row, each character being defined by a 6×8 bit matrix. In this case an attribute table is not required; instead the character colour is defined by a colour register on the 9929 itself that stores one foreground and one background colour. The character shapes are held in RAM in the VDP's memory

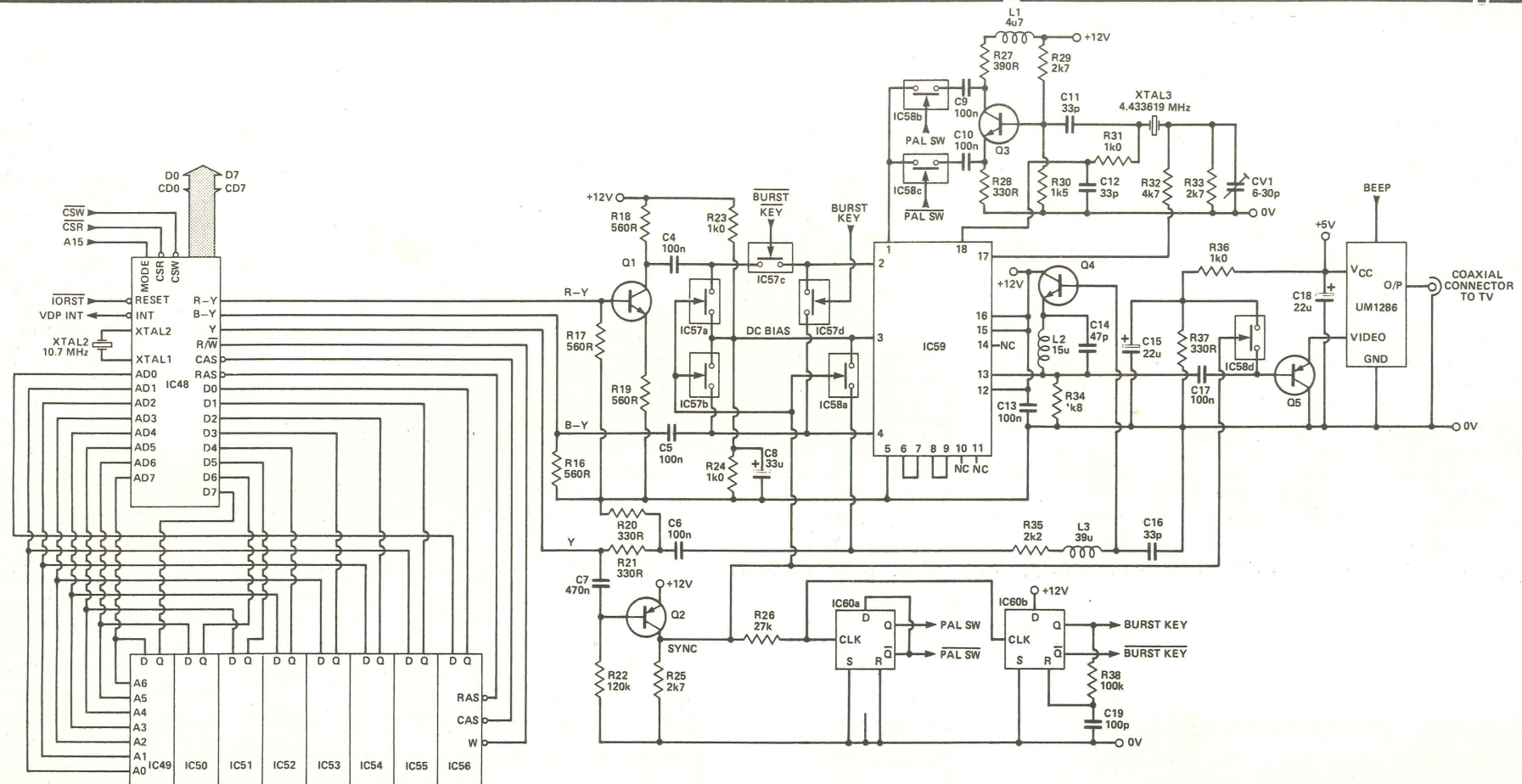


Fig. 4 Circuit diagram of the video display circuitry.

HOW IT WORKS — VIDEO DISPLAY PROCESSOR AND PAL ENCODER

The TMS9929 (IC48) is a 625-line non-interlaced video display processor; it directly drives 16K of memory which is completely separate from the main CPU memory. The VDP fetches data from its DRAM (ICs 49-56) at such a rate that the DRAM is automatically refreshed many times over. There's very little else to say about this section of the circuitry — IC48 does everything internally! The VDP outputs a composite luminance and sync waveform on the Y output and colour difference signals on the R-Y and B-Y outputs. These signals contain all

the information needed to produce either R-G-B or PAL-encoded colour. The VDP is controlled by a two-byte memory-mapped slot in high memory.

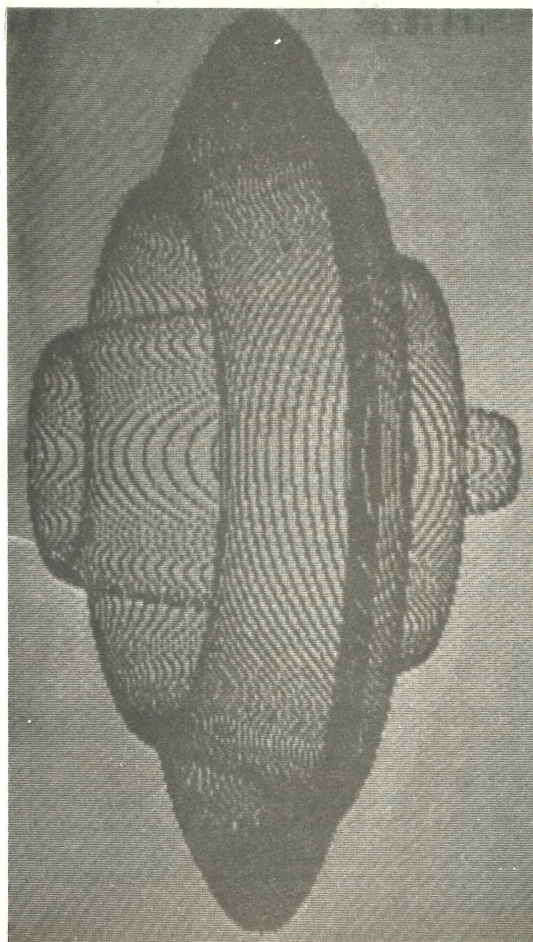
The sync is separated from the Y (luminance) signal by Q2 and associated circuitry, and used to drive DC restoration clamps IC57a,b,58a; these charge capacitors C4, 5, 6 to a reference voltage during the sync pulse. The sync is also used to toggle the PAL (Phase Alternate Line) switch, IC60a, which gates an inverted or non-inverted chroma oscillator

signal into one of the two analogue multipliers in IC59. The chroma oscillator is built around Q3 and XTAL3, a crystal whose resonant frequency is 4.433619MHz, that of the PAL colour subcarrier. The non-inverted signal is taken from the collector of Q3 via C9 and IC58b, while the inverted signal is taken from Q3 emitter via C10 and IC58c.

The second flip-flop, IC60b, is used as a monostable which, at the beginning of each line, connects the burst pulse which occurs on the B-Y signal to the

R-Y input of IC59. This switching is done by IC57d. The inverting amplifier Q1 on the R-Y line from the VDP, IC48, is to match the direction of the burst pulse with the direction of the video signal to yield the correct colours.

The luminance signal is low-pass-filtered by R28, L3, C16 and then summed with the chrominance output of IC59 via the chroma trap L2, C14; this filter removes colour fringing effects. The signal is then DC-shifted by another DC restoration clamp (IC58d) to feed the RF modulator.



and so may be easily user-modified. The CHAR command allows any of the 256 possible character definitions to be altered.

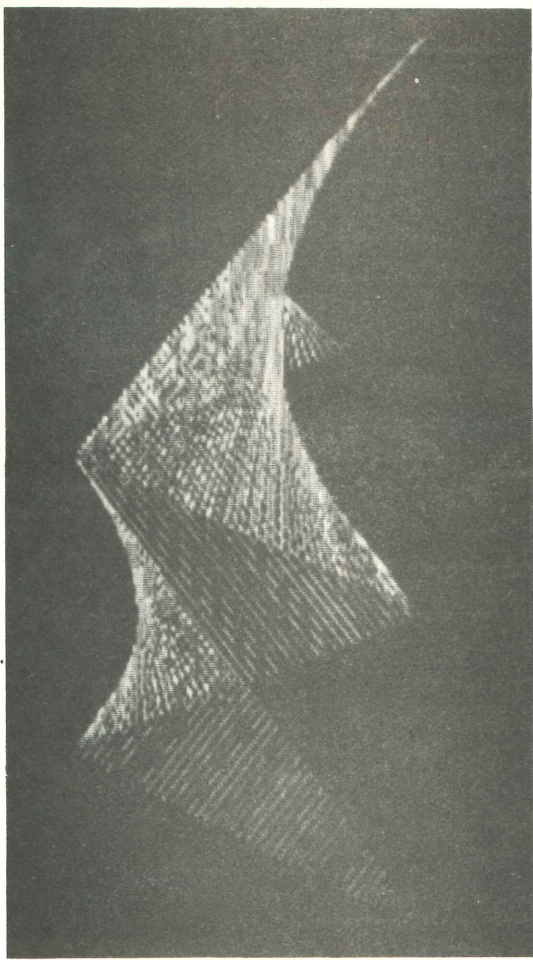
Table 1 shows the 16 colours which are available; this 'palette' has been arranged to give not only a good colour display, but also a good monochrome display, as the colours produce an even grey scale on a black-and-white TV. Eight grey levels are generated.

One peculiarity may have caught your eye in Table 1: what is the point of a transparent colour? A transparent object will allow you to see what's behind it, but in most graphic displays 'behind' is meaningless. However, the VDP in the Cortex considers its display to consist of 36 planes prioritised one

above the other. When you look at the screen you're seeing an image which can be considered analogous to holding 36 colour slides, one above the other in a stack, and peering through them all.

The rearmost plane is black to allow images to be built up over it. The next plane is for external video and need not concern us here. On top of this is the backdrop plane which lies directly behind the text/graphic plane. This defines the border colour as well. Since this plane defines the colour of the whole screen, it is now obvious that the only way to see the external video input or the black rearmost plane is to set the backdrop to transparent. The text/graphic plane is written to by the TEXT and GRAPH commands discussed earlier.

This leaves another 32 planes sitting in front of the four mentioned above; these are called the sprite planes. A sprite is a graphic shape that can be user-defined from BASIC with the SPRITE command. Sprites can be displayed in a variety of sizes depending on the size and magnification flags; these give four



possible modes. SIZE 0 means that a block of 8x8 bits is used to define the sprite, while SIZE 1 uses 16x16 bits (but reduces the total number of different shapes from 256 to 64).

The display size can be varied with the MAG command; MAG 0 maps one bit in the shape onto one pixel while MAG 1 maps one bit onto a 2x2 block of pixels on the screen.

Each sprite has four attributes associated with it; its plane (or priority), its colour and its X and Y screen coordinates. Again, each sprite can be one of 16 colours, and those bits set to 1 in the sprite definition adopt the defined colour while the other bits are set to transparent. The screen coordinates define the position of the top left-hand corner of the sprite, and sprite positions can therefore be rapidly changed by simply altering two

bytes in memory. The colour can be changed equally quickly by altering one byte. Because the planes are prioritised, 0 to 31, if any shape is positioned coincident with another shape, only the one with the highest priority plane will be displayed. This gives rise to simple 3D simulation. A status flag is set to indicate when

any two sprites 'touch' each other. Any point in the text or graphic plane will only be seen if all the points directly above it on the 32 sprite planes are transparent.

All these features mean you can generate very versatile and complex displays; but they also use up a reasonable amount of memory. We don't believe that screen RAM should be stolen from the user RAM, so the VDP only occupies two bytes in the CPU memory map. These two registers are all that's required to write all the relevant information through the VDP chip and into its own 16K of DRAM.

More next month!

BUYLINES

Powertran are supplying complete kits of parts and component packs for the Cortex. A complete 64K Cortex kit will cost £295 plus VAT, carriage free. A ready-built 64K Cortex will cost £395 plus VAT, carriage free. Prices for additions (eg floppy discs, RS232C interface, memory expansion etc) and for component packs (eg PCB, semiconductors etc) can be found in Powertran's brochure. Powertran Cybernetics, Portway Industrial Estate, Andover, Hants SP10 2NM. Telephone 0264 64455.

TABLE 1

Code	Colour	Code	Colour
0	transparent	8	medium red
1	black	9	light red
2	medium green	10	dark yellow
3	light green	11	light yellow
4	dark blue	12	dark green
5	light blue	13	magenta
6	dark red	14	grey
7	cyan	15	white